PROJECT 02 | DATA SCIENCE & AI

Data Visualization with Matplotlib & Seaborn

Beginner | About 6 hours | Data Science & AI

WHAT YOU WILL BUILD

Create informative charts including histograms, heatmaps, box plots, and correlation matrices from real datasets.

Create a set of clear, honest charts from a real dataset, including histograms, box plots, scatter plots and a correlation heatmap.

YOUR REQUIREMENTS

- Bar charts and scatter plots
- Correlation heatmap generation
- Custom color palettes and plot formatting

WHAT YOU WILL LEARN

- Choosing the right chart for the question
- Making charts with matplotlib and seaborn
- Labelling, colouring and formatting for clarity
- Writing a one-line takeaway for every chart

MATERIALS YOU NEED

TOOL	**Python 3.10+ with Jupyter**	Install Python from python.org, then pip install jupyter pandas numpy matplotlib seaborn scikit-learn. Google Colab is a free alternative that needs no install.
DATA	**Tips dataset**	Load with sns.load_dataset('tips'). It has total bill, tip, day, time, size and smoker columns.
READING	**Seaborn tutorial**	The official tutorial at seaborn.pydata.org explains each plot type with examples.
READING	**From Data to Viz**	The website data-to-viz.com helps you choose the right chart type.

STEP-BY-STEP GUIDE

1 Choose the questions

Write four questions, for example How are bills distributed? Do tips differ by day? Does a bigger bill mean a bigger tip? Which numbers move together?

2 Match charts to questions

Use a histogram for one number, a box plot to compare groups, a scatter plot for two numbers and a heatmap for correlations.

3 Draw the first charts

Make each chart with seaborn, then add a title, axis labels and units. Increase the figure size so labels are readable.

4 Use colour carefully

Pick one palette, use colour to mean something (for example time of day) and avoid red and green together for colour-blind readers.

5 Add the correlation heatmap

Compute df.select_dtypes('number').corr() and plot it with annotations. Write what the strongest relationship is.

6 Combine into a dashboard image

Place four charts in a 2 by 2 grid with plt.subplots and save at 150 dpi.

7 Write the takeaways

Under each chart write one sentence saying what it shows, such as Most bills are between 10 and 25 dollars.

8 Review for honesty

Check your axes start at sensible values, nothing is cut off and every chart can be understood without explanation.

WORKED EXAMPLE OF A FINISHED PROJECT

Four charts in one figure

A complete working example to adapt.

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")
sns.set_theme(style="whitegrid", palette="deep")
fig, axes = plt.subplots(2, 2, figsize=(11, 8))

sns.histplot(tips["total_bill"], bins=20, ax=axes[0, 0])
```

```
axes[0, 0].set(title="Most bills are under 25 dollars", xlabel="Total bill (USD)",
ylabel="Number of meals")

sns.boxplot(data=tips, x="day", y="tip", ax=axes[0, 1])
axes[0, 1].set(title="Tips by day", xlabel="Day", ylabel="Tip (USD)")

sns.scatterplot(data=tips, x="total_bill", y="tip", hue="time", ax=axes[1, 0])
axes[1, 0].set(title="Bigger bills bring bigger tips", xlabel="Total bill (USD)", ylabel="Tip
(USD)")

sns.heatmap(tips.select_dtypes("number").corr(), annot=True, cmap="Blues", ax=axes[1, 1])
axes[1, 1].set_title("Correlation between numbers")

plt.tight_layout()
plt.savefig("tips_overview.png", dpi=150)
```

CHECK YOUR WORK BEFORE YOU SUBMIT

- [ ] Each chart type fits its question
- [ ] Every chart has a title, labelled axes and units
- [ ] Colours are consistent and colour-blind friendly
- [ ] The heatmap is annotated and explained
- [ ] A takeaway sentence sits under every chart
- [ ] The final image is clear at normal size

WHAT TO SUBMIT

- A link to your notebook or code
- The saved chart image(s)
- Your written takeaways

You submit a link. Put your files in Google Drive, GitHub, Figma, YouTube or similar, allow anyone with the link to view them, and paste the link when you submit.

COMMON MISTAKES TO AVOID

- Avoid 3D and pie charts for comparing many values.
- A chart title should state the finding, not just the variables.
- Do not hide the zero line on bar charts.
- Use plt.tight_layout() so labels are not cut off.

EXPECTED OUTCOME

Learn to tell compelling stories through visual data graphics.

Submit your work from the My Tasks page of your dashboard. A mentor reviews it and sends feedback; if changes are requested you can update and resubmit. Write your own solution - the example above shows the standard to aim for, not something to copy.